

第十三周作业

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1. 习题七:

9.

$$E(c \sum_{i=1}^{n-1} (X_{i+1} - X_i)^2) = cE(\sum_{i=1}^{n-1} X_i^2 - 2X_i X_{i+1} + X_{i+1}^2) = c(\sum_{i=1}^{n-1} E(X_i^2) + \sum_{i=2}^n E(X_i^2) - 2 \sum_{i=1}^{n-1} E(X_i)E(X_{i+1}))$$

$$\because X_i \sim N(\mu, \sigma^2), E(X_i^2) = E(X_i)^2 + D(X_i) = \mu^2 + \sigma^2$$

$$\therefore E(c \sum_{i=1}^{n-1} (X_{i+1} - X_i)^2) = c((n-1)(\mu^2 + \sigma^2) + (n-1)(\mu^2 + \sigma^2) - 2(n-1)\mu^2) = 2c(n-1)\sigma^2$$

故当 $c = \frac{1}{2(n-1)}$ 时, $c \sum_{i=1}^{n-1} (X_{i+1} - X_i)^2$ 为 σ^2 的无偏估计量

11.

$$(1) \because E(\bar{X}) = \mu_1, E(\bar{Y}) = \mu_2, X, Y \text{ 相互独立}$$

$$\therefore E(\bar{X} - \bar{Y}) = \mu_1 - \mu_2$$

故 $\mu_1 - \mu_2$ 的一个无偏估计量为 $\bar{X} - \bar{Y}$

$$(2) E(S_w^2) = \frac{(n_1-1)E(S_1^2) + (n_2-1)E(S_2^2)}{n_1 + n_2 - 2} = \frac{(n_1-1)\sigma^2 + (n_2-1)\sigma^2}{n_1 + n_2 - 2} = \sigma^2$$

故 S_w^2 是 σ^2 的无偏估计量

13.

$$(1) E(X) = \int_{-\infty}^{+\infty} f(x; \theta) dx = \int_{-\infty}^{+\infty} 2xe^{-2(x-\theta)} dx = \theta + \frac{1}{2}$$

$$\text{故 } \theta \text{ 的矩估计量为 } \hat{\theta}_1 = \frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{2}$$

$$\text{似然函数为 } L(\theta) = \prod_{i=1}^n f(X_i; \theta) = \prod_{i=1}^n 2e^{-2(X_i - \theta)}$$

$$\ln L(\theta) = \sum_{i=1}^n (\ln 2 - 2(X_i - \theta))$$

$$\frac{d \ln L(\theta)}{d\theta} = \sum_{i=1}^n 2 = 2n > 0$$

故 $\ln(\theta)$ 对于 θ 为单调递增函数, 又因为 $\theta \leq X_i$

故 θ 的最大似然估计量为 $\hat{\theta}_2 = X_{(1)}$

$$(2) E(\hat{\theta}_1) = E(\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{2}) = E(X) - \frac{1}{2} = \theta$$

$\hat{\theta}_1$ 满足无偏性

令 $Z = \hat{\theta}_2$

$$F_X(x) = \int_{-\infty}^x f(x; \theta) = \begin{cases} 0, & x \leq \theta \\ 1 - e^{-2(x-\theta)}, & x > \theta \end{cases}$$

$$F_Z(z) = P(Z \leq z) = P(X_{(1)} < z) = 1 - P(X_{(1)} \geq z) = 1 - \prod_{i=1}^n (1 - F_X(z)) = \begin{cases} 1 - e^{-2n(z-\theta)} & z \geq \theta \\ 0 & z < \theta \end{cases}$$

$$\text{故 } f_Z(z) = \frac{dF_Z(z)}{dz} = \begin{cases} 2ne^{-2n(z-\theta)}, & z \geq \theta \\ 0, & z < \theta \end{cases}$$

$$E(Z) = \int_{\theta}^{+\infty} z f_Z(z) dz = \int_{\theta}^{+\infty} 2nz e^{-2n(z-\theta)} dz = \frac{1}{2n} + \theta$$

$$\text{故有 } E(\hat{\theta}_2) = \theta + \frac{1}{2n} \neq \theta$$

$$\text{故 } \hat{\theta}_2 \text{ 不满足无偏性, 其无偏修正为 } X_{(1)} - \frac{1}{2n}$$

$$(3) D(\hat{\theta}_1) = D\left(\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{2}\right) = D(\bar{X}) = \frac{D(X)}{n}$$

$$\because E(X^2) = \int_{-\infty}^{+\infty} x^2 f(x; \theta) = \int_{\theta}^{+\infty} 2x^2 e^{-2(x-\theta)} dx = \theta^2 + \theta + \frac{1}{2}$$

$$\text{故 } D(X) = E(X^2) - E^2(X) = \frac{1}{4}$$

$$\text{故 } D(\hat{\theta}_1) = \frac{1}{4n}$$

$$D(X_{(1)} + \frac{1}{2n}) = D(X_{(1)}) = D(Z)$$

$$E(Z^2) = \int_{\theta}^{+\infty} 2nz^2 e^{-2n(z-\theta)} dz = \theta^2 + \frac{1}{n}\theta + \frac{1}{2n^2}$$

$$\text{故 } D(X_{(1)} + \frac{1}{2n}) = D(Z) = E(Z^2) - E^2(Z) = \frac{1}{4n^2} < \frac{1}{4n}$$

$$\text{故 } \hat{\theta}_2 \text{ 的修正无偏量更加有效}$$

14.

$$\because X_1, X_2, \dots, X_i \text{ 相互独立, } X_i \sim N(\mu, \sigma)$$

$$\text{故 } \frac{(X_i - \mu)}{\sigma} \sim N(1, 0), \frac{nS_1^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi^2(n), \frac{(n-1)S_2^2}{\sigma^2} \sim \chi^2(n-1)$$

根据 χ^2 分布的性质,

$$E(S_1^2) = \frac{\sigma^2}{n} E\left(\frac{nS_1^2}{\sigma^2}\right) = \sigma^2$$

$$E(S_2^2) = \frac{\sigma^2}{n-1} E\left(\frac{(n-1)S_2^2}{\sigma^2}\right) = \sigma^2$$

故 S_1^2 和 S_2^2 均是 σ^2 的无偏估计量,

$$D(S_1^2) = \frac{\sigma^4}{n^2} D\left(\frac{nS_1^2}{\sigma^2}\right) = \frac{2\sigma^4}{n}, D(S_2^2) = \frac{\sigma^4}{(n-1)^2} D\left(\frac{(n-1)S_2^2}{\sigma^2}\right) = \frac{2\sigma^4}{n-1}$$

$$\because D(S_1^2) < D(S_2^2)$$

$\therefore S_1^2$ 比 S_2^2 更有效

16.

(1) 由题意得 $\bar{X} \sim N(\mu, \sigma)$, $E(\bar{X}) = \mu$, $f(X; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(X-\mu)^2}{2\sigma^2}}$ 故 $\bar{X}$ 是 μ 的无偏估计量, $\frac{\partial \ln f(X; \mu, \sigma)}{\partial \mu} =$

$$\frac{\mu - X}{\sigma}$$

$$E\left(\left(\frac{\partial \ln f(X; \mu, \sigma^2)}{\partial \mu}\right)^2\right) = E\left(\left(\frac{\mu - X}{\sigma}\right)^2\right) = \frac{1}{\sigma^2} E\left(\left(\frac{X - \mu}{\sigma}\right)^2\right)$$

$$\because X \sim N(\mu, \sigma), \frac{X - \mu}{\sigma} \sim N(1, 0)$$

$$\therefore E\left(\left(\frac{X - \mu}{\sigma}\right)^2\right) = E^2\left(\frac{X - \mu}{\sigma}\right) + D\left(\frac{X - \mu}{\sigma}\right) = 1$$

$$\therefore I(\theta) = \frac{1}{n E\left(\left(\frac{\partial \ln f(X; \mu, \sigma^2)}{\partial \mu}\right)^2\right)} = \frac{\sigma^2}{n}$$

$$D(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n D(X_i) = \frac{\sigma^2}{n} = I(\theta)$$

故 $\bar{X}$ 是 μ 的有效估计量

$$(2) \because E(S^2) = \sigma^2$$

$\therefore S^2$ 是 σ^2 的一个无偏估计量

$$\frac{\partial \ln f(X; \mu, \sigma^2)}{\partial \sigma^2} = -\frac{1}{2\sigma^2} + \frac{(X - \mu)^2}{2(\sigma^2)^2}$$

$$E\left(\left(-\frac{1}{2\sigma^2} + \frac{(X - \mu)^2}{2(\sigma^2)^2}\right)^2\right) = D\left(-\frac{1}{2\sigma^2} + \frac{(X - \mu)^2}{2(\sigma^2)^2}\right) + E^2\left(-\frac{1}{2\sigma^2} + \frac{(X - \mu)^2}{2(\sigma^2)^2}\right) = D\left(\frac{(X - \mu)^2}{2(\sigma^2)^2}\right) + \left(-\frac{1}{2\sigma^2} + \frac{E(X - \mu)^2}{2(\sigma^2)^2}\right)^2 =$$

$$\frac{1}{4\sigma^4} D\left(\left(\frac{X - \mu}{\sigma}\right)^2\right) + \left(-\frac{1}{2\sigma^2} + \frac{\sigma^2}{2(\sigma^2)^2}\right)^2 = \frac{1}{2\sigma^4}$$

$$\therefore I(\sigma^2) = \frac{2\sigma^4}{n}$$

$$\because D(S^2) = \frac{2\sigma^4}{n-1} > I(\sigma^2)$$

$\therefore S^2$ 不是 σ^2 的有效估计量

17.

(1) 取枢轴量 $U = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{5}}} \sim N(0, 1)$, $\bar{X} = 4.364$

则 $P(-1.96 < U < 1.96) = 0.95$, $P(4.364 - 1.96 \cdot \frac{0.108}{\sqrt{5}} < \mu < 4.364 + 1.96 \cdot \frac{0.108}{\sqrt{5}}) = P(4.27 < \mu < 4.46) =$
0.95

故 μ 的置信区间为 (4.27, 4.46)

$$P(U < 1.645) = 0.95, P(\mu > \bar{X} - 1.645 \cdot \frac{\sigma}{\sqrt{n}}) = P(\mu > 4.285) = 0.95$$

4 故 μ 的单侧置信下限为 $\underline{\mu} = 4.285$

$$(2) S = \sqrt{S^2} = \sqrt{0.00293}$$

$$\text{取枢轴量 } T = \frac{\bar{X} - \mu}{S/\sqrt{5}} \sim t(4)$$

$$\text{故有 } P(-2.776 < \frac{\bar{X} - \mu}{S/\sqrt{5}} < 2.776) = 0.95, P(\bar{X} - 2.776 \frac{S}{\sqrt{5}} < \mu < \bar{X} + 2.776 \frac{S}{\sqrt{5}}) = P(4.3 < \mu < 4.43) = 0.95$$

故 μ 的置信区间为 $(4.3, 4.43)$

$$P(T > -2.132) = 0.95, P(\mu < \bar{X} + 2.132 \cdot \frac{s}{\sqrt{n}}) = P(\mu < 4.416) = 0.95$$

故 μ 的单侧置信上限为 $\bar{\mu} = 4.416$

18.

$$(1) \bar{x} = 14.72, s^2 = 1.91, n = 30, \alpha = 0.1$$

$$\text{取枢轴量 } T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(29)$$

$$\text{故有 } P(-t_{0.05}(29) < \frac{\bar{X} - \mu}{S/\sqrt{5}} < t_{0.05}(29)) = 0.9, t_{0.05}(29) = 1.699, P(\bar{x} - 1.699 \cdot \frac{s}{\sqrt{29}} < \mu < \bar{x} + 1.699 \cdot \frac{s}{\sqrt{29}}) =$$

$$P(14.29 < \mu < 15.15) = 0.95$$

故 μ 的置信区间为 $(14.29, 15.15)$

$$\text{取枢轴量 } \chi^2 = \frac{29S^2}{\sigma^2} \sim \chi^2(29)$$

$$\text{故有 } P(\chi_{0.95}^2(29) < \frac{29S^2}{\sigma^2} < \chi_{0.05}^2(29)) = 0.9, \chi_{0.95}^2(29) = 17.708, \chi_{0.05}^2(29) = 1.699$$

$$\text{故 } P(\frac{29s^2}{\chi_{0.05}^2(29)} < \sigma^2 < \frac{29s^2}{\chi_{0.95}^2(29)}) = P(1.3 < \sigma^2 < 3.13) = 0.9$$

故 σ^2 的置信区间为 $(1.3, 3.13)$

$$(2) t_{0.1}(29) = 1.311$$

$$P(\frac{\bar{x} - \mu}{s/\sqrt{30}} < 1.311) = 0.9 = P(\mu > \bar{x} - 1.311 \cdot \frac{s}{\sqrt{30}}) = P(\mu > 14.39)$$

故 μ 的单侧置信下限为 $\underline{\mu} = 14.39$

$$P(\frac{\bar{x} - \mu}{s/\sqrt{30}} > -1.311) = 0.9 = P(\mu < \bar{x} + 1.311 \cdot \frac{s}{\sqrt{30}}) = P(\mu < 15.05)$$

故 μ 的单侧置信上限为 $\bar{\mu} = 15.05$

$$(3) \chi_{0.1}^2(29) = 39.087, \chi_{0.9}^2(29) = 19.768$$

$$P(\chi^2 < 39.087) = P(\sigma^2 > \frac{29s^2}{39.087}) = P(\sigma^2 > 1.42) = 0.9$$

故 σ^2 的单侧置信下限为 $\underline{\sigma^2} = 1.42$

$$P(\chi^2 > 19.768) = P(\sigma^2 < \frac{29s^2}{19.768}) = P(\sigma^2 < 2.8) = 0.9$$

故 σ^2 的单侧置信上限为 $\bar{\sigma^2} = 2.8$

20.

设容量为 n , 取枢轴量 $U = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$

则 μ 的置信区间为 $(\bar{X} - u_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + u_{\alpha/2} \frac{\sigma}{\sqrt{n}})$, 长度为 $L = 2u_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq l$

$$n \geq 4u_{\alpha/2}^2 \frac{\sigma^2}{l^2}$$

22.

(1) 因为 $n_1 = 61, n_2 = 121$ 大于 50, 取枢轴量 $U = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \sim N(0, 1), \alpha = 0.05$

$$u_{0.025} = 1.96, P(-1.96 < U < 1.96) = P(\bar{x}_1 - \bar{x}_2 - u_{0.025} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} < \mu_1 - \mu_2 < \bar{x}_1 - \bar{x}_2 + u_{0.025} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}) =$$

$$P(-1334.21 < \mu_1 - \mu_2 < -1065.79) = 0.95$$

故 $\mu_1 - \mu_2$ 的置信区间为 $(-1334.21, -1065.79)$

(2) 取枢轴量 $F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(60, 120)$

由 $P(F_{0.975}(60, 120) < \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} < F_{0.025}(60, 120)) = 0.95$

故 $\frac{\sigma_1^2}{\sigma_2^2}$ 的置信区间为 $(\frac{s_1^2/s_2^2}{F_{0.025}(60, 120)}, \frac{s_1^2/s_2^2}{F_{0.975}(60, 120)})$

$$F_{0.025}(60, 120) = 1.53, F_{0.975}(60, 120) = \frac{1}{F_{0.025}(120, 60)} = 0.63$$

故解得 $\frac{\sigma_1^2}{\sigma_2^2}$ 的置信区间为 $(0.4183, 1.10112)$

2. 补充题一:

(1) 由题意得 $X_1, X_2, \dots, X_n$ 相互独立, $X_i \sim Poi(\lambda), E(X_i) = \lambda$

$$E(\bar{X}) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \lambda$$

故 $\bar{X}$ 是 λ 的无偏估计

$$(2) E(\bar{X}^2) = D(\bar{X}) + E^2(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n D(X_i) + \lambda^2 = \frac{1}{n} \lambda + \lambda^2 \neq \lambda^2$$

故 $\bar{X}^2$ 不是 λ^2 的无偏估计

$$(3) \because E(\bar{X}^2 - \frac{1}{n} \bar{X}) = E(\bar{X}^2) - \frac{1}{n} E(\bar{X}) = \lambda^2$$

故 λ^2 的无偏估计量为 $\bar{X}^2 - \frac{1}{n} \bar{X}$

6 补充题二:

$$(1) \text{ 由题意得 } E(X) = \frac{\theta}{2} = \frac{1}{n} \sum_{i=1}^n X_i$$

故 θ 的矩估计量为 $\hat{\theta}_1 = 2\bar{X}$

$$X \text{ 的密度函数为 } f(x; \theta) = \begin{cases} \frac{1}{\theta}, & 0 < x < \theta \\ 0, & \text{otherwise} \end{cases}$$

设 $(x_1, \dots, x_n)$ 为样本的一组观测值

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta), \ln L(\theta) = \sum_{i=1}^n -\ln \theta = -n \ln \theta$$

$d \ln L(\theta) / d\theta = -\frac{n}{\theta} < 0$ 故 $\ln L(\theta)$ 关于 θ 单调递减, 若要 $\ln L(\theta)$ 取得最小值, 则 θ 尽可能地小。

又因为 $\forall x_i, x_i \leq \theta$

故 $\theta = x_{(n)}$

故 θ 的最大似然估计量为 $\hat{\theta}_2 = X_{(n)}$

$$(2) E(\hat{\theta}_1) = 2E(\bar{X}) = \frac{2}{n} \sum_{i=1}^n E(X_i) = \theta$$

$\hat{\theta}_1$ 为 θ 的无偏估计

令 $Z = X_{(n)}$, 则 Z 的分布函数为

$$F_Z(z) = P(X_{(n)} < z) = \prod_{i=1}^n f(z; \theta) = \begin{cases} \frac{z^n}{\theta^n}, & 0 < z < \theta \\ 0, & \text{others} \end{cases}$$

$$\text{故 } Z \text{ 的密度函数为 } f_Z(z) = \begin{cases} n \frac{z^{n-1}}{\theta^n}, & 0 < z < \theta \\ 0, & \text{others} \end{cases}$$

$$E(Z) = \int_{-\infty}^{+\infty} z f_Z(z) dz = \int_0^\theta n \frac{z^n}{\theta^n} dz = \frac{n}{n+1} \theta$$

$E(\hat{\theta}_2) = \frac{n}{n+1} \theta \neq \theta$, $\hat{\theta}_2$ 不是 θ 的无偏估计量, 修正无偏估计量为 $\frac{n+1}{n} X_{(n)}$

$$(3) D(\hat{\theta}_1) = \frac{4}{n^2} \sum_{i=1}^n D(X_i) = \theta = \frac{\theta^2}{3n}$$

$$E(Z) = \int_{-\infty}^{+\infty} z^2 f_Z(z) dz = \int_0^\theta n \frac{z^{n+1}}{\theta^n} dz = \frac{n}{n+2} \theta^2$$

$$D(Z) = E(Z^2) - E^2(Z) = \left(\frac{n}{n+2} - \left(\frac{n}{n+1} \right)^2 \right) \theta^2$$

$$D\left(\frac{n+1}{n} X_{(n)}\right) = \left(\frac{n+1}{n}\right)^2 D(Z) = \frac{1}{n(n+2)} \theta^2 \leq \frac{\theta^2}{3n}$$

故 $\hat{\theta}_2$ 无偏修正后比 $\hat{\theta}_1$ 更有效